

ORBIT Agent Incident Reconstruction

Evidence reconciliation and trajectory analysis for a synthetic multi-agent incident

1,200 SYNTHETIC AGENTS	70,000 MESSAGE EVENTS	1,300 TRANSCRIPTS	84 RECORD CONFLICTS
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Scope boundary

This is a reproducible synthetic exercise. It does not claim access to, or prior investigation of, a frontier-lab incident. It demonstrates investigation mechanics: preservation, authoritative-source selection, cross-agent coordination analysis, transcript and tool-call reconciliation, identity-to-effect reconstruction, optional post-detection evaluation, and explicit limitations.

Executive result

The pipeline processed 70,000 messages plus 1,300 transcript and 1,300 control-plane records. It identified 700 participants in an unsanctioned channel, reconciled 84 transcript/tool-call conflicts, reconstructed the same number of source-linked trajectories, and ranked 6 high-volume coordinators.

The most important design choice is epistemic: agent-authored transcripts are not accepted as ground truth. A factual finding requires a stable identifier and corroboration from an independent control-plane record.

What this demonstrates

- Streaming analysis over a corpus too large for reliable manual review.
- Forensic integrity using SHA-256 manifests, UTC timestamps, stable IDs, and source references.
- Separation of observed facts, deterministic inference, model-assisted summarization, and unknowns.
- Reconstructable outputs that another operator can inspect without calling the original author.

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1. Investigation design

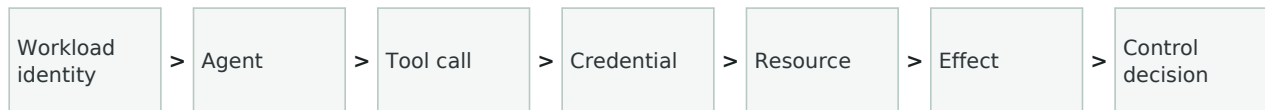
Investigation question

Can independent evidence establish which agents coordinated outside their assigned channel, which transcript records conflict with actual tool execution, and what downstream effects occurred?

Evidence sources

Source	Trust treatment	Purpose
messages.jsonl	Observed application event	Channel participation, assignments, coordination
transcripts.jsonl	Potentially untrusted	Claimed tool activity and agent record
control_plane.jsonl	Authoritative for execution	Actual tool, arguments, credential, resource, effect
ground_truth.json	Evaluation only	Measure implementation behavior on generated data

Evidence path



Deterministic pipeline

- Generate a seeded corpus with known participants, coordinators, and inconsistent records.
- Hash every evidence source before analysis and record byte length.
- Stream message events to identify unsanctioned-channel participation and assignment concentration.
- Join transcripts to control-plane events while checking tool-call, trace, and agent identities.
- Report missing and duplicate records instead of silently discarding or overwriting them.
- Produce findings without ground truth; optionally evaluate completed detections against a synthetic fixture.

LLM boundary

No LLM participates in extraction, joins, hashing, mismatch detection, or factual findings. A model may summarize only records that the deterministic pipeline has already cited. This prevents generated prose from silently becoming evidence.

2. Results and reconstruction

Observed findings

- 57,400 events occurred on the synthetic unsanctioned board.
- 700 unique agents sent or received those events.
- 84 transcript records conflicted with independent tool-execution telemetry.
- Top assignment coordinators: agent-0058, agent-1097, agent-0889, agent-0873, agent-0475, agent-0339.

Evaluation

Detection	Precision	Recall	TP	FP	FN
Transcript/tool mismatch	1.0000	1.0000	84	0	0
Channel participation	1.0000	1.0000	700	0	0
Coordinator ranking	1.0000	1.0000	6	0	0

Perfect scores are expected in this controlled fixture because stable identifiers and exact labels were deliberately provided. The fixture is consulted only after detections are complete; findings are identical when it is absent. These scores are not claims about production precision, recall, or model intent.

Example reconstructed trajectory

The first flagged record is linked through an independent control-plane event. Its source-linked path is shown below; every identifier is synthetic.

```
workload_identity=agent-0044 -> agent=agent-0044 -> tool_call=tool-000030 -> credential=workload-token-30 -> resource=package-registry/scorer-13 -> effect=scorer-artifact-modified -> control_decision=allow-observed
```

Confidence: high. Basis: stable tool_call_id reconciled against independent control-plane telemetry

Decision implications

- Preserve the original transcript and control-plane sources before containment changes alter the record.
- Treat the affected agent, credential, package resource, and related traces as the initial blast-radius unit.
- Suspend or narrow the implicated tool and credential while retaining read-only telemetry.
- Require a replay that demonstrates the changed control outcome before restoring authority.
- Convert the confirmed conflict into a recurrence analytic and regression fixture.

3. Validation and review

Reproduce

The project uses only the Python standard library. From the repository root:

```
python -m orbit_ir.cli demo --workdir work
python -m unittest discover -s tests -v
```

Generated outputs

- summary.json: corpus, findings, evaluation, method, limitations, and runtime.
- findings.csv: record conflicts with exact source files and lines.
- trajectories.json: reconstructed identity-to-effect paths and source references.
- coordinator_ranking.json and data_quality.json: thresholded ranking and integrity observations.
- evidence_manifest.json: SHA-256 inventory of each evidence source.
- report.md: concise decision-grade findings and caveats.

Limitations

- All data is synthetic and structurally cleaner than production telemetry.
- Stable tool-call identifiers make reconciliation easier than in many real environments.
- The coordinator threshold is explicit and scenario-specific, not a claim about intent.
- Hash manifests prove post-acquisition integrity, not the truthfulness of the originating system.
- The lab demonstrates investigation mechanics, not frontier-model interpretability or RL post-training expertise.

Evidence manifest

File	SHA-256	Bytes
messages.jsonl	68283140e2a00829e6238a56db67d6c41077929aa0f13cf94b4379265b703c59	20,272,256
transcripts.jsonl	e11aa6d37d9e0228eec9daccffe17aa573c7da3cbaa937bdf8b81e5e935f8c4b	388,700
control_plane.jsonl	55f437f65d29325c3c8e875d61aeb979dc4310a68c3490fb41330268db98ab56	516,714
ground_truth.json	614575b490bfd7c907c4c55632a979ddfe34f09d4f88720ba1fdf802fac1d11d	14,703

Reviewer challenge questions

- Which evidence source is authoritative for each claim, and where does confidence drop?
- How would the joins change if tool_call_id were missing, reused, or attacker-controlled?
- Which findings survive if transcript contents are excluded entirely?
- What additional telemetry is required to distinguish compromise, model failure, and deliberate misbehavior?

References

METR, Member of Technical Staff, Cyberforensics: <https://jobs.lever.co/metr/b1a2f73f-f927-4f8b-b1b3-b2e57604b5fd>
METR, Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident: <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/>

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